



22CH203 – ENGINEERING CHEMISTRY II

UNIT – III & METAL CORROSION AND ITS PREVENTION

Oxidation of metals: Electrochemical origin of corrosion – electromigration - electron transfer in the presence and absence of moisture - galvanic series. Strategies for corrosion control: Galvanic anode and impressed current.

LESSON PLAN 2:

Electron transfer in the presence of moisture - galvanic series.

TOPIC - TYPES OF ELECTROCHEMICAL CORROSION

1. GALVANIC CORROSION

When two different metals are in contact with each other in presence of an aqueous solution or moisture, galvanic corrosion occurs. Between the metals, the more active metal (with more negative electrode potential) acts as anode and the less active metal (with less negative electrode potential) acts as cathode.

1.1 Examples

i) Zn-Fe couple

In case of Zn-Fe couple, zinc (more active or higher in emf series) oxidizes in preference to iron (less active metal) i.e., Zn acts as anode and undergoes corrosion and Fe acts as cathode.

ii) Cu-Fe couple

In case of Fe-Cu couple, in which iron (more active, when compared to Cu) dissolves in preference to copper (less active) i.e., Fe acts as anode and undergoes corrosion and Cu acts as cathode.

iii) Steel screw in a brass marine hardware corrodes: This is due to galvanic corrosion. Iron (higher position in electrochemical series) becomes anodic and is attacked and corroded, while brass (lower in electrochemical series) acts as cathodic and is not attacked.

iv) Bolt and Nut made of the same metal is preferred: It is preferred in practice, because galvanic corrosion is avoided due to homogeneous metals (no anodic and cathodic part).

1.2 Prevention: Galvanic corrosion can be minimized by providing an insulating material between the two metals.

1.3 Galvanic Series

The **galvanic series** (or **electropotential series**) determines the nobility of metals and semi-metals. When two metals are submerged in an electrolyte, while also electrically connected by some

external conductor, the less noble (base) will experience galvanic corrosion. The rate of corrosion is determined by the electrolyte and the difference in nobility. The difference can be measured as a difference in voltage potential: the less noble metal is the one with a lower (that is, more negative) electrode potential than the nobler one, and will function as the anode (electron or anion attractor) within the electrolyte device functioning as described above (a galvanic cell). Galvanic reaction is the principle upon which batteries are based.



2. Differential aeration (or) concentration cell corrosion:

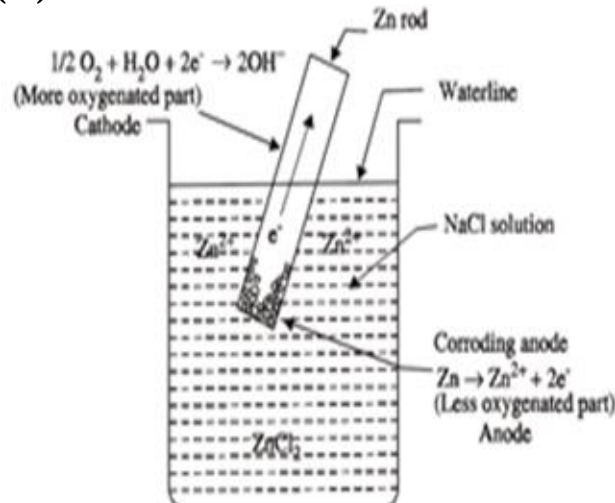


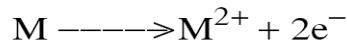
Figure 1.1 Differential aeration corrosion

This type of corrosion occurs when a metal is exposed to varying concentration of oxygen or any electrolyte on the surface of the base metal

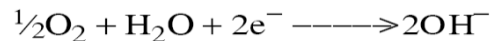
Example: Metals partially immersed in water (or) conducting solution (called water line corrosion).

If a metal is partially immersed in a conducting solution (Fig. 1.1) the metal part above the solution is more aerated and hence become *cathodic*. On the other hand, the metal part inside the solution is less aerated and thus, become *anodic* and suffers corrosion.

At Anode: Corrosion Occurs



At Cathode: OH^{-} Ions are produced **Reduction**



Examples of differential aeration corrosion

(a) Pitting or localised corrosion.

(b) Water line corrosion

(c) Pipeline corrosion

2.1 Pitting corrosion

Pitting is a localised attack, resulting in the formation of a hole around which the metal is relatively not attacked. Let us consider a drop of water or aqueous NaCl resting on a metal surface (Fig. 1.2). The area covered by the drop of water acts as an anode due to less oxygen concentration and suffers corrosion. The uncovered area (freely exposed to air) acts as a cathode due to high oxygen concentration. The rate of corrosion will be more, when the area of cathode is larger and the area of anode is smaller. Therefore, more and more material is removed from the same spot. Thus a small hole or pit is formed on the surface of the metal.

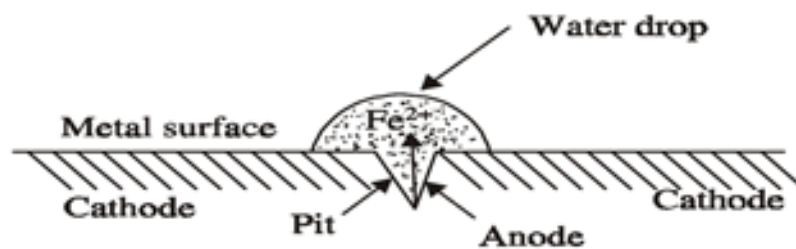
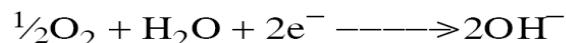


Figure 1.2 Pitting corrosion

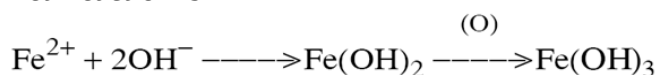
At Anode: Iron is oxidized to Fe^{2+} ions



At Cathode: Oxygen is converted to OH^{-} ions.



Net Reaction is



This type of intense corrosion is called pitting.

2.2 Water line corrosion

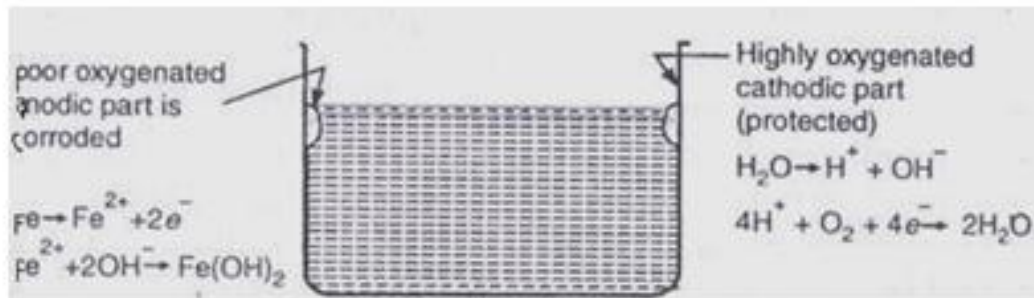


Figure 1.3 Waterline corrosion

Waterline corrosion is a case of differential aeration corrosion (Fig. 1.3), more prevalent in cases such as ocean going ships, water storage steel tanks, etc., in which a portion of the metal is always under water.

The waterline corrosion takes place due to the formation of differential oxygen concentration cells. The part of the metal below the water line is exposed only to dissolved oxygen while the part above the water is exposed to higher oxygen concentration of the atmosphere. Thus, part of the metal below the water acts as anode and undergoes corrosion and part above the waterline is free from corrosion. A distinct brown line is formed just below the water line due to the deposition of rust.

2.3 Pipeline corrosion

Differential aeration corrosion may also occur in different parts of pipeline. Buried pipelines or cables passing from one type of soil to another say, from clay (less aerated) to sand (more aerated) may get corroded due to differential aeration as shown in fig 1.4.

Other examples for differential aeration corrosion

1. Corrosion occurring under metal washers, where oxygen cannot diffuse easily.
2. Lead pipeline passing through clay to cinders undergo corrosion. Since the pipeline under cinders is more aerated, it gets corroded easily.

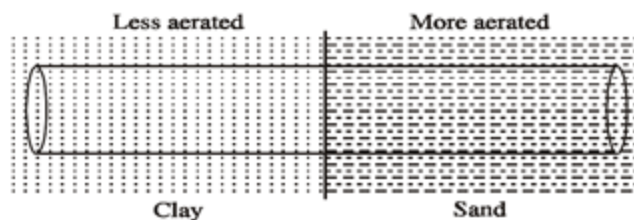


Figure 1.4 Pipeline corrosion

Discussion Questions:

1. Compare the monometallic and bimetallic corrosion based on the electrode potential.
2. Predict the corroding part of metal when it is partially immersed in an electrolyte.